Practical 12

1. Aim

To install, configure and use the Zabbix monitoring tool on an Ubuntu 20.04.6 virtual machine running in VMware Workstation Pro, and monitor system performance parameters such as CPU utilization, memory usage, disk space and network activity.

2. Requirements

Hardware and software requirements:

Component	Specification
Virtualization software	VMware Workstation Pro
Guest OS	Ubuntu 20.04.6 LTS
Processor	2 vCPUs
RAM	4 GB minimum
Hard disk	30 GB
Monitoring tool	Zabbix 4.0 LTS
Database	MySQL
Web server	Apache
Browser	Firefox / Chrome

3. Theory

VM performance monitoring is the process of observing and analyzing the performance of a virtual machine to ensure efficient operation.

Zabbix is an open-source monitoring tool used to monitor servers, virtual machines, network devices and applications.

It collects performance data and displays it through a web-based dashboard.

Main features of Zabbix:

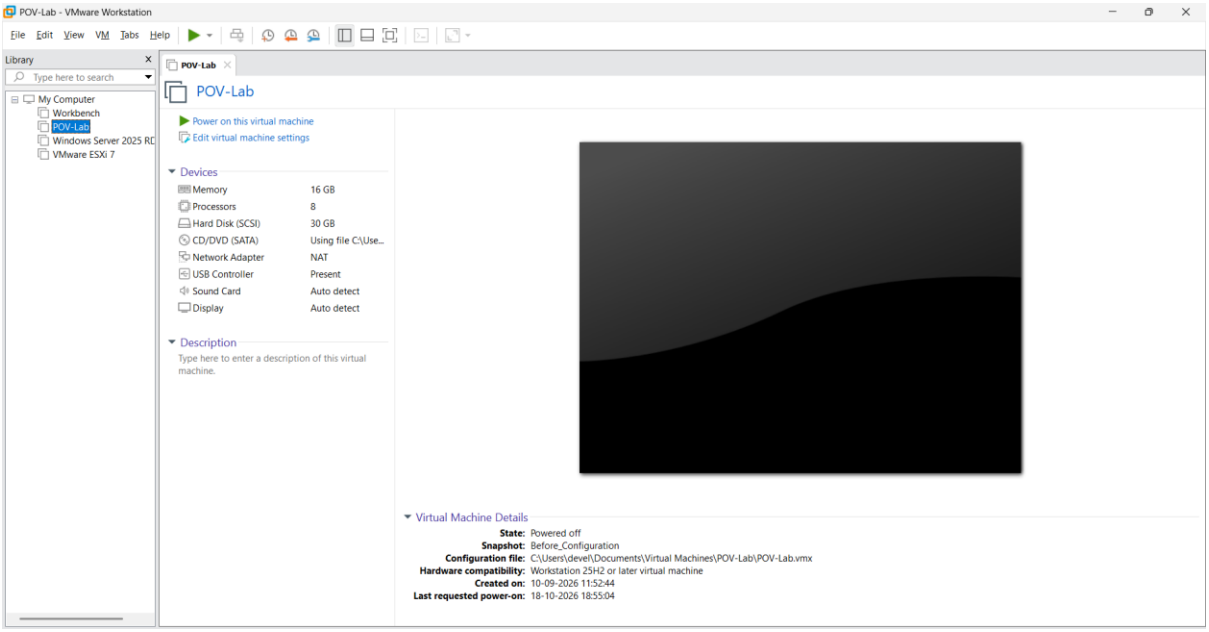
1. CPU utilization monitoring.
2. Memory usage monitoring.
3. Disk space monitoring.
4. Network traffic monitoring.
5. System uptime monitoring.
6. Graphs, alerts and performance history.

Zabbix uses a server-agent architecture. The Zabbix Agent collects performance information from the Ubuntu VM, while the Zabbix Server processes and stores the collected information.

4. Procedure

Step 1: Start Ubuntu Virtual Machine

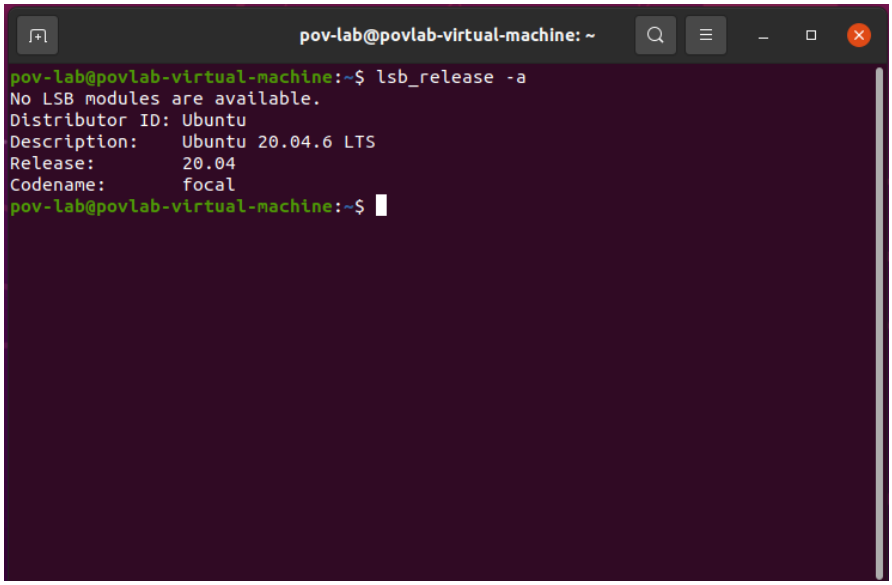
1. Open VMware Workstation Pro.
2. Select the Ubuntu 20.04.6 virtual machine.



3. Click Power On.
4. Log in to Ubuntu.

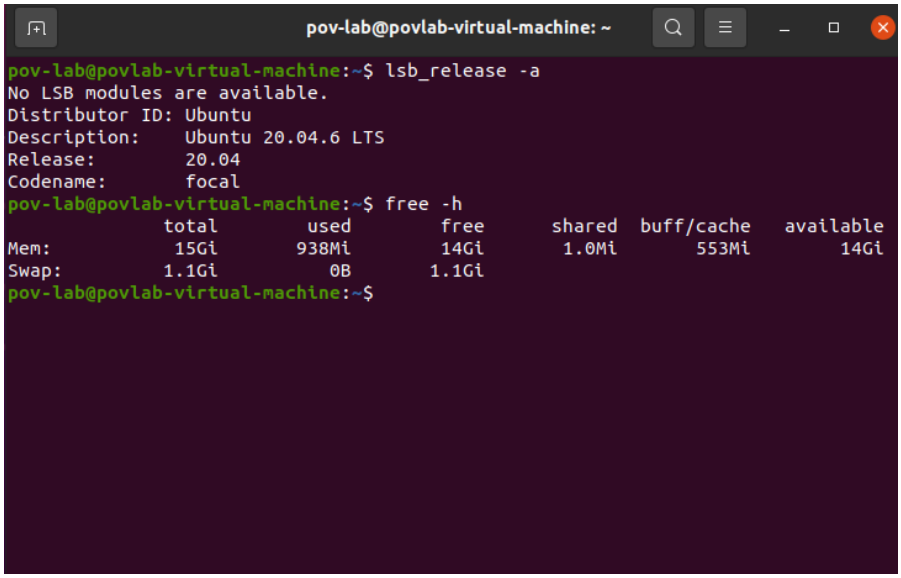
Verify the operating system:

lsb_release -a



Check available memory:

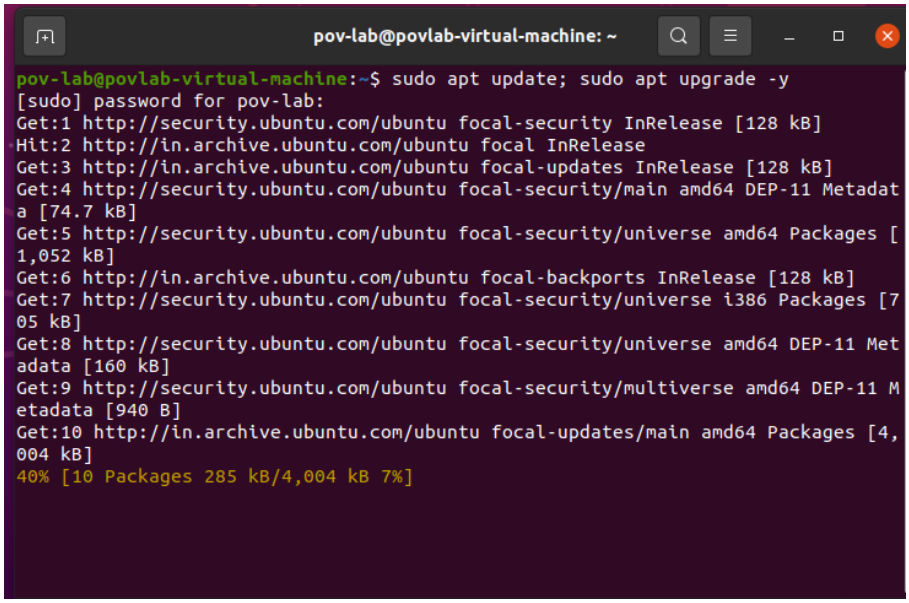
free -h



Update the system:

sudo apt update

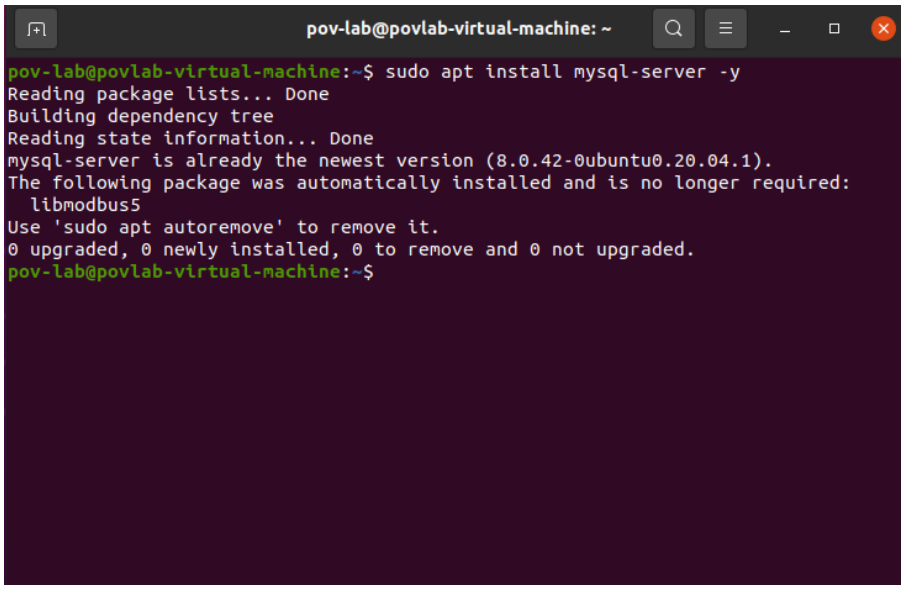
sudo apt upgrade -y



Step 2: Install MySQL Server (If not already installed) Update your package index and install the MySQL server.

sudo apt update

sudo apt install mysql-server -y



Step 3: Create the Zabbix Database and User Create the database, set up the zabbix user with your lab password, and grant the necessary permissions.

sudo mysql -e "CREATE DATABASE zabbix character set utf8 collate utf8_bin;"

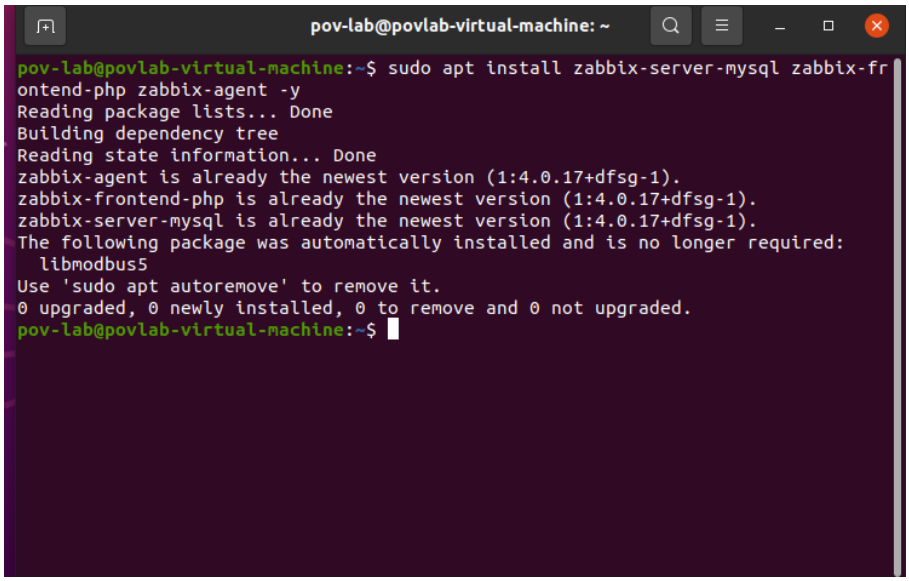
sudo mysql -e "CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'LabPassword@2026';"

sudo mysql -e "GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';"

sudo mysql -e "FLUSH PRIVILEGES;"

Step 4: Install Zabbix Server, Web Frontend, and Agent Install the native Ubuntu 20.04 Zabbix packages (which defaults to version 4.0.17).

sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-agent -y



Step 5: Import the Zabbix Database Schema and Data Import the three required SQL files in this exact order to build the database. When prompted for a password for each command, enter LabPassword@2026.

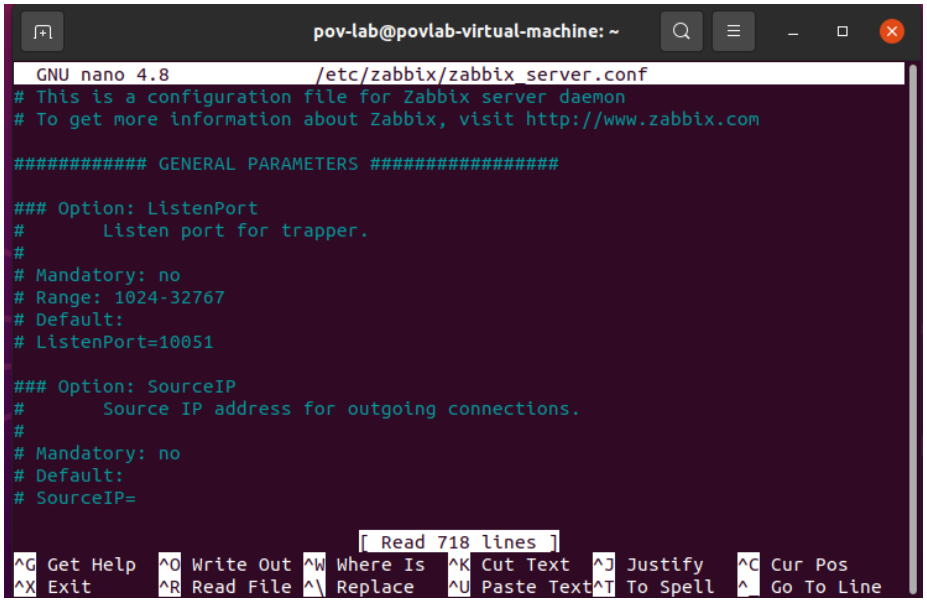
zcat /usr/share/zabbix-server-mysql/schema.sql.gz | mysql -uzabbix -p zabbix

zcat /usr/share/zabbix-server-mysql/images.sql.gz | mysql -uzabbix -p zabbix

zcat /usr/share/zabbix-server-mysql/data.sql.gz | mysql -uzabbix -p zabbix

Step 6: Configure the Zabbix Server Database Password Link your Zabbix server to the MySQL database by adding the password to the configuration file.

sudo nano /etc/zabbix/zabbix_server.conf



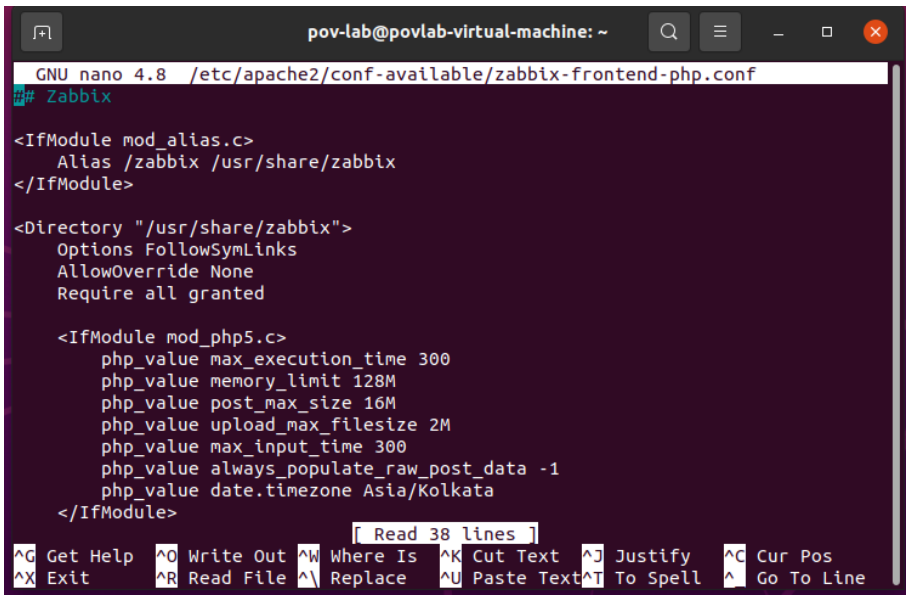
Locate the line # DBPassword= (around line 124), remove the #, and set it to your lab password:

DBPassword=LabPassword@2026

Save and exit (Ctrl + O, Enter, Ctrl + X).

Step 7: Configure the Apache Web Server Timezone Set the correct PHP timezone for the Zabbix web interface.

sudo nano /etc/apache2/conf-available/zabbix-frontend-php.conf



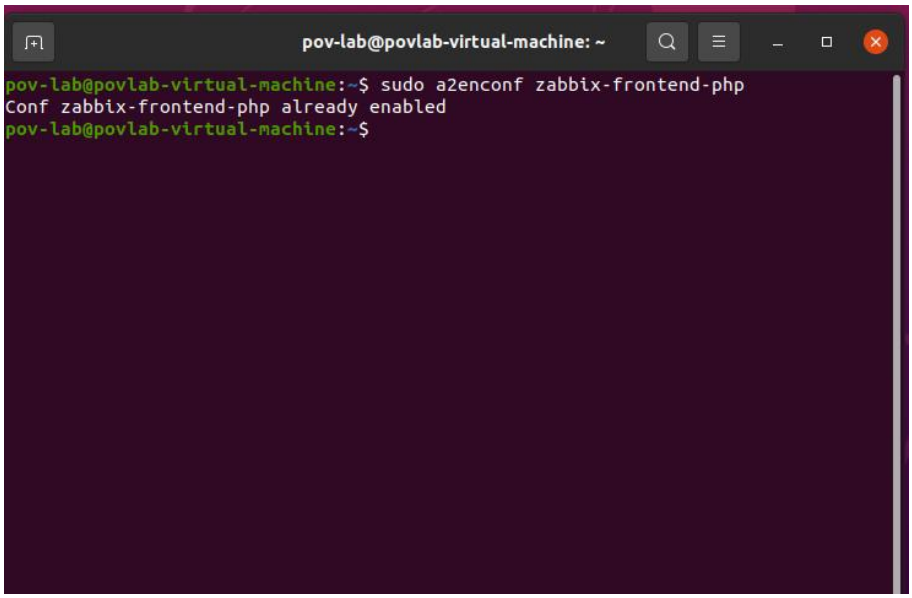
Locate the timezone line (e.g., # php_value date.timezone Europe/Riga), remove the #, and update it to your local time:

php_value date.timezone Asia/Kolkata

Save and exit (Ctrl + O, Enter, Ctrl + X).

Step 8: Enable the Apache Configuration Activate the Zabbix frontend web alias in Apache so the browser can find the files.

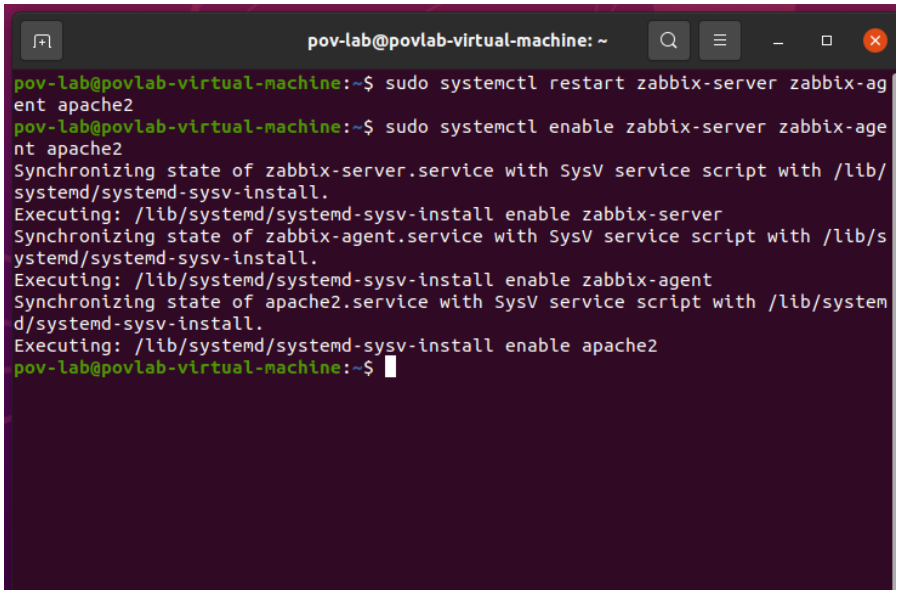
sudo a2enconf zabbix-frontend-php



Step 9: Start and Enable All Services Restart all relevant services to apply the configurations, and set them to start automatically on system boot.

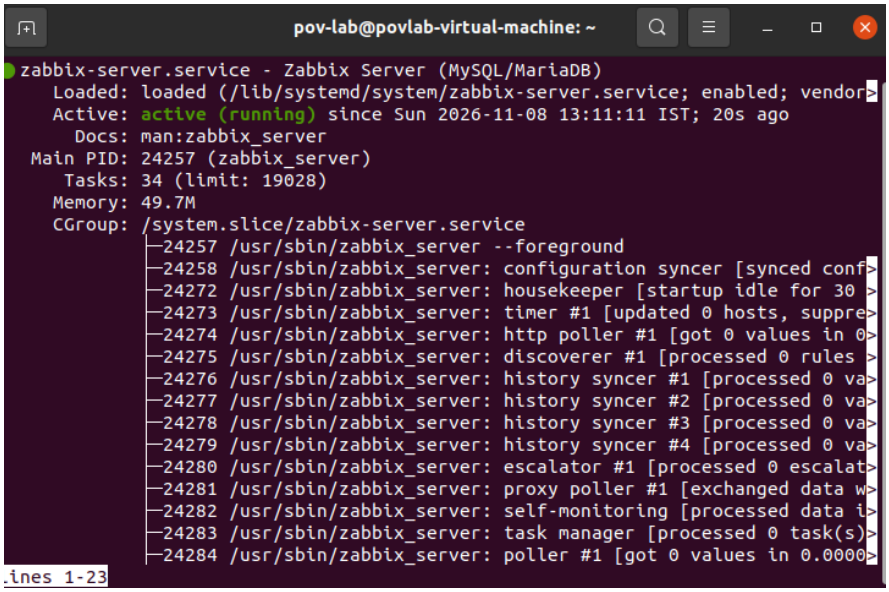
sudo systemctl restart zabbix-server zabbix-agent apache2

sudo systemctl enable zabbix-server zabbix-agent apache2



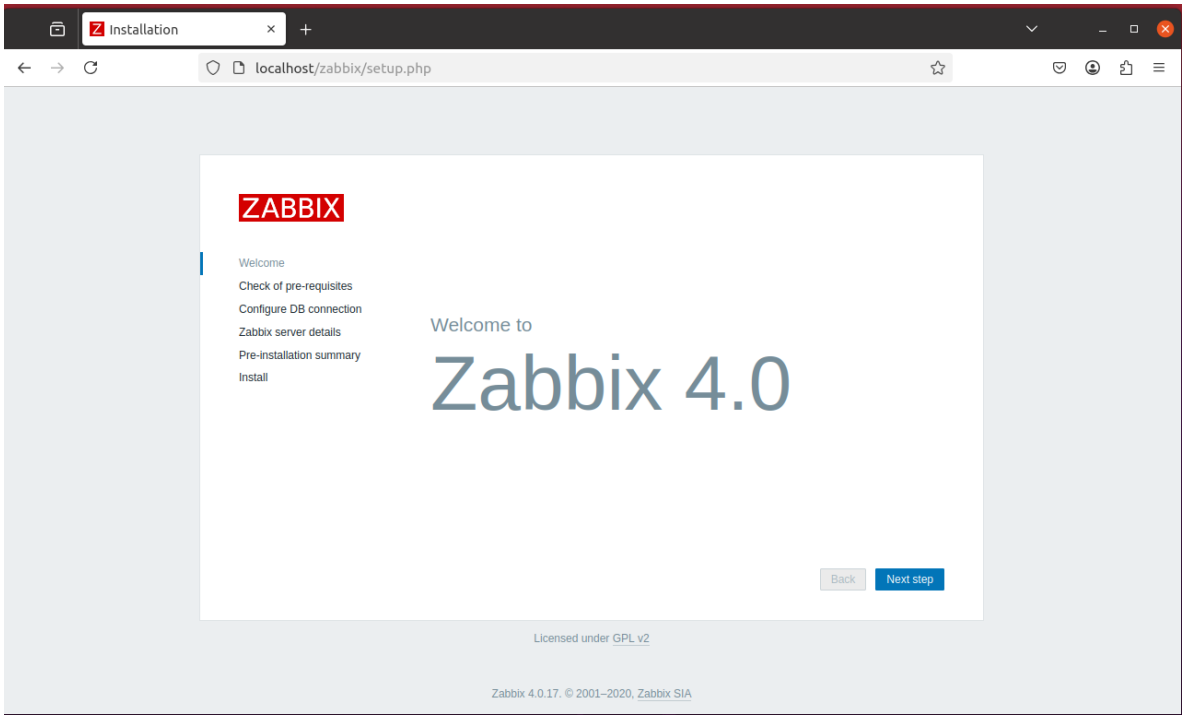
Check the status to confirm the Zabbix server is running without errors:

sudo systemctl status zabbix-server

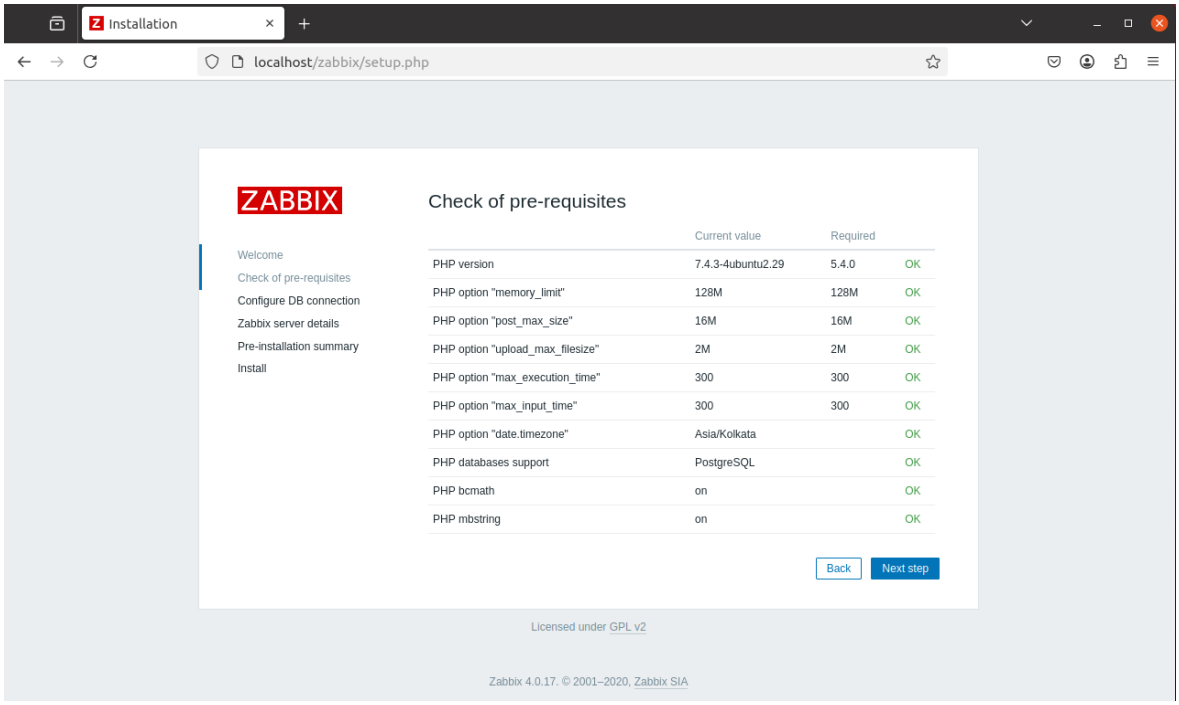


Step 10: Complete the Web Interface Setup

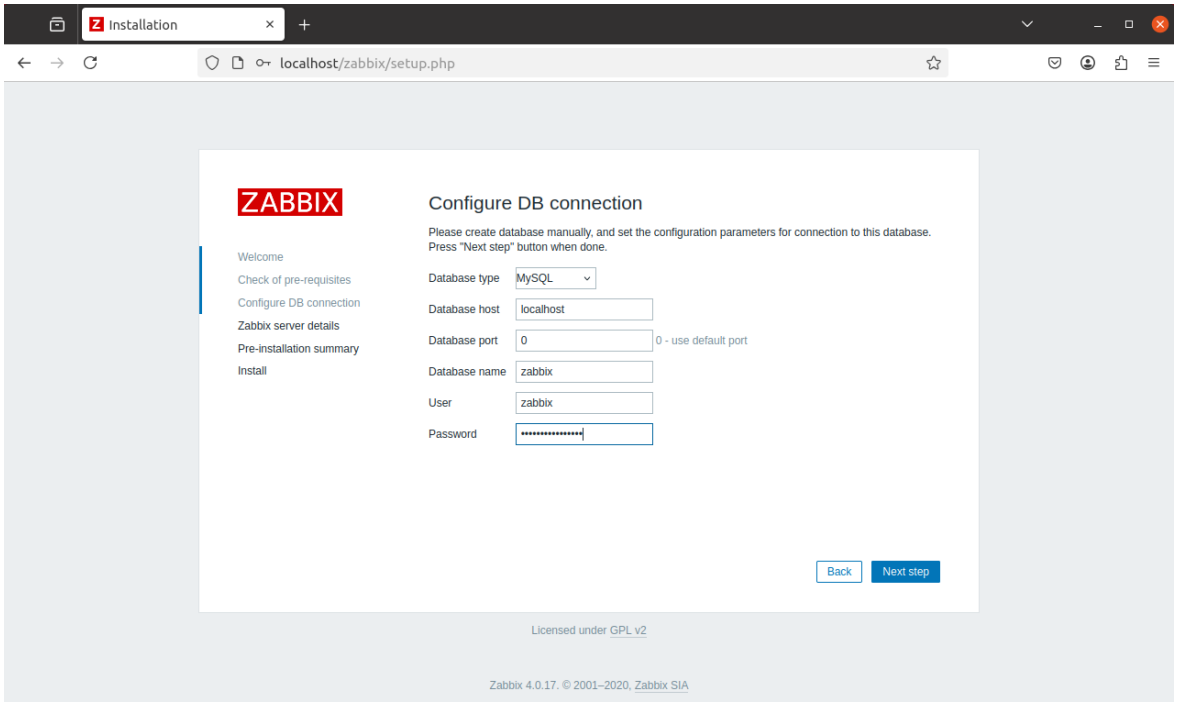
1. Open a web browser and navigate to http://localhost/zabbix.
2. On the Zabbix 4.0 welcome screen, click **Next step**.

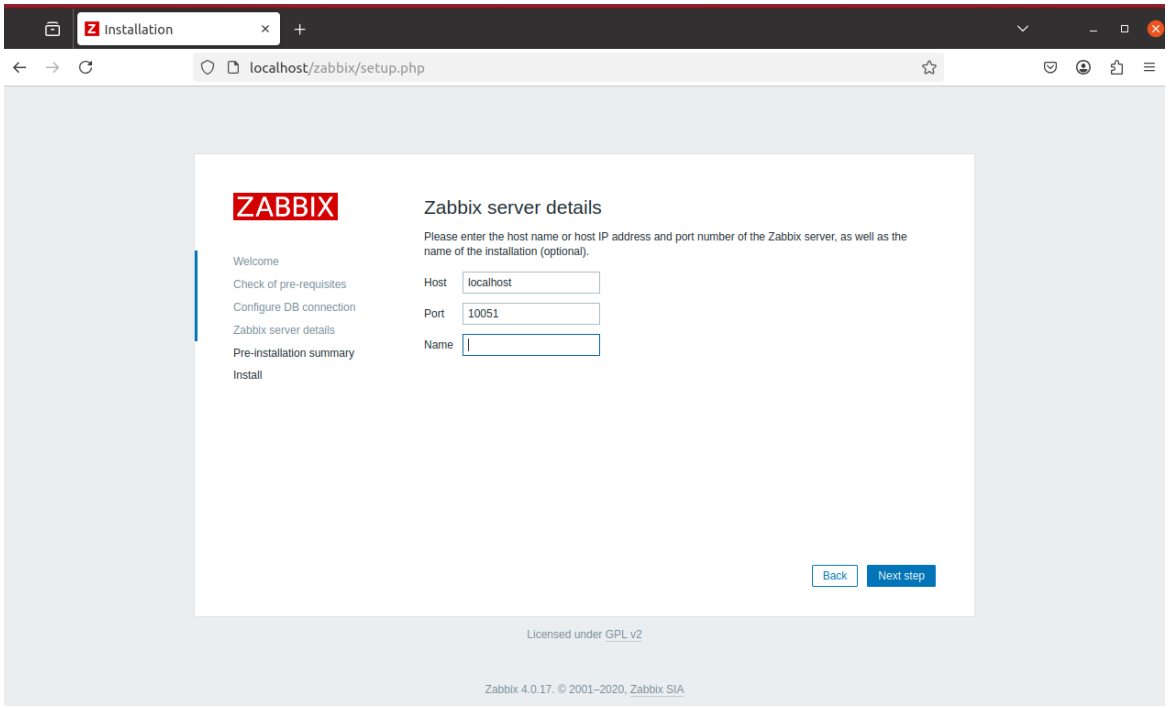


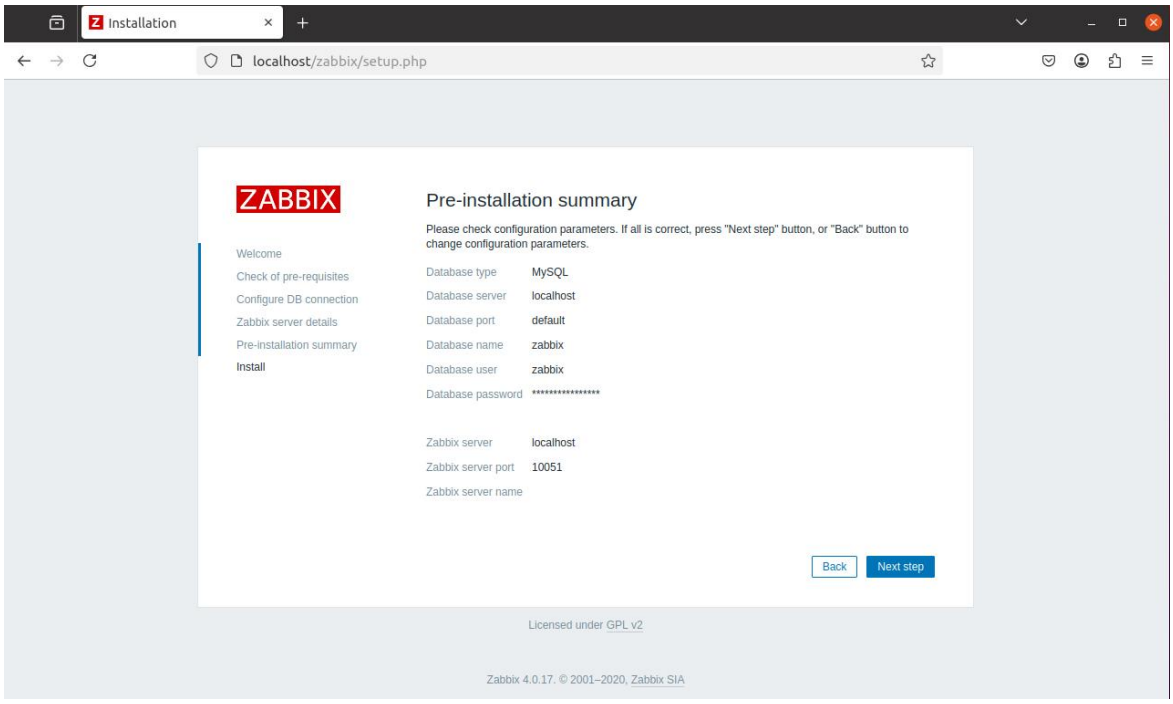
3. Verify that all pre-requisites are marked as OK and proceed.



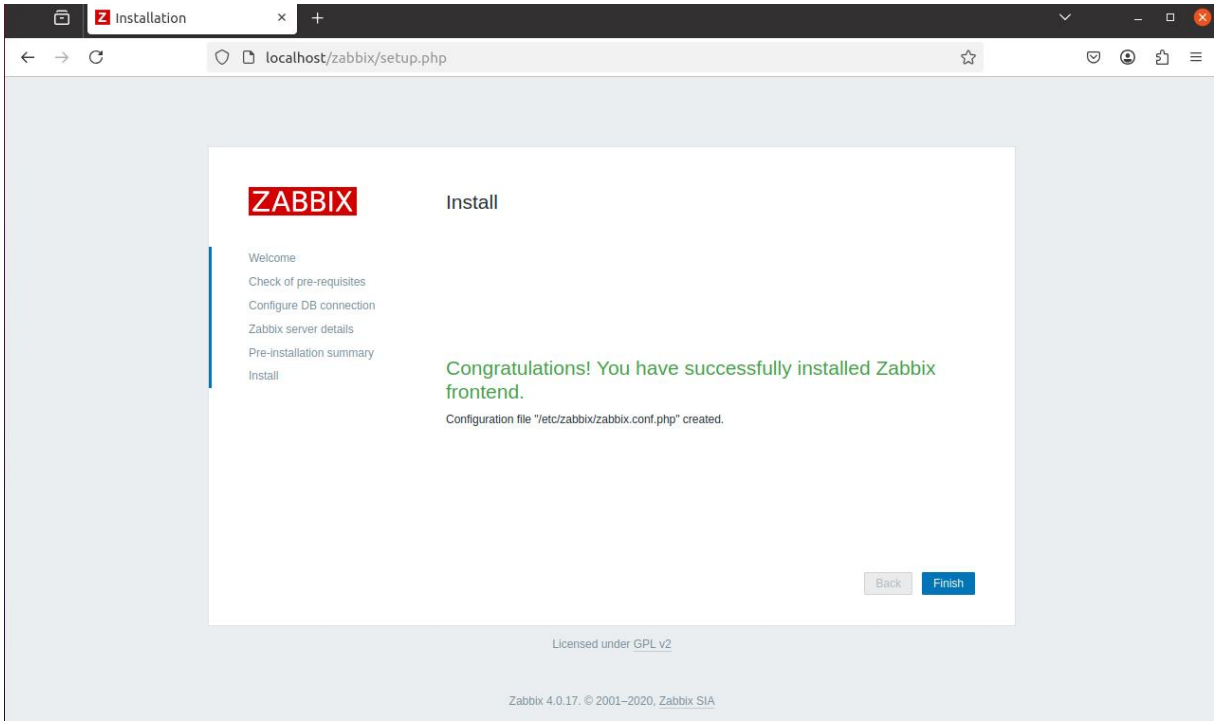
4. On the **Configure DB connection** screen, enter the user as **zabbix** & password: LabPassword@2026.



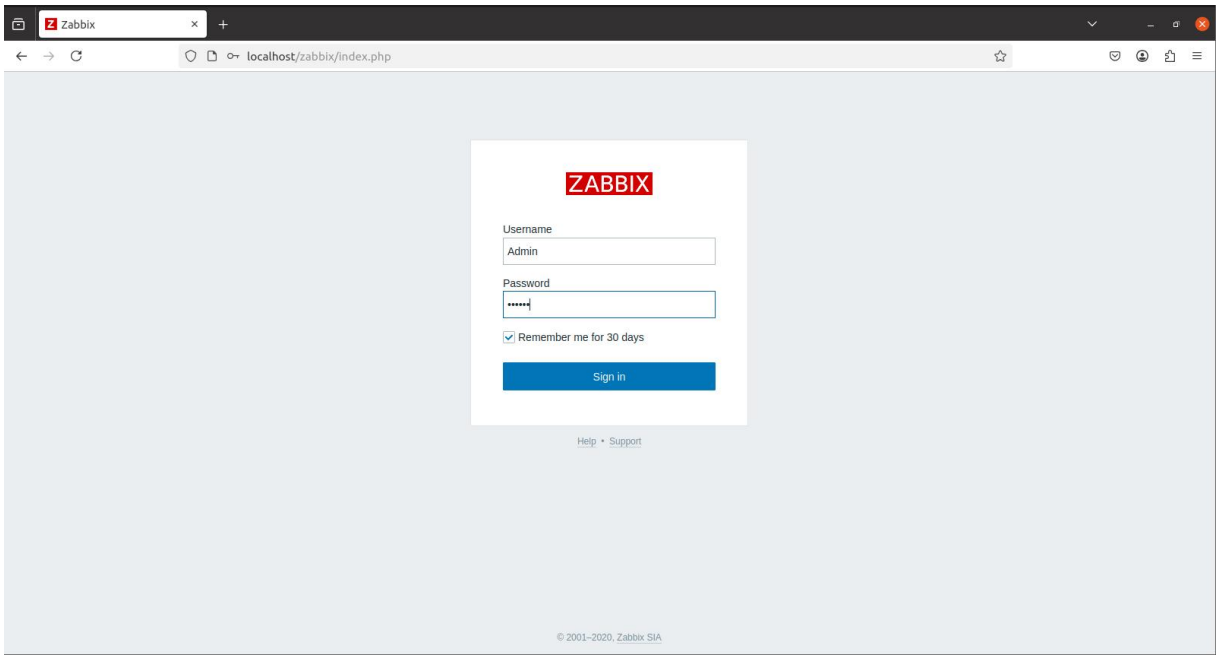


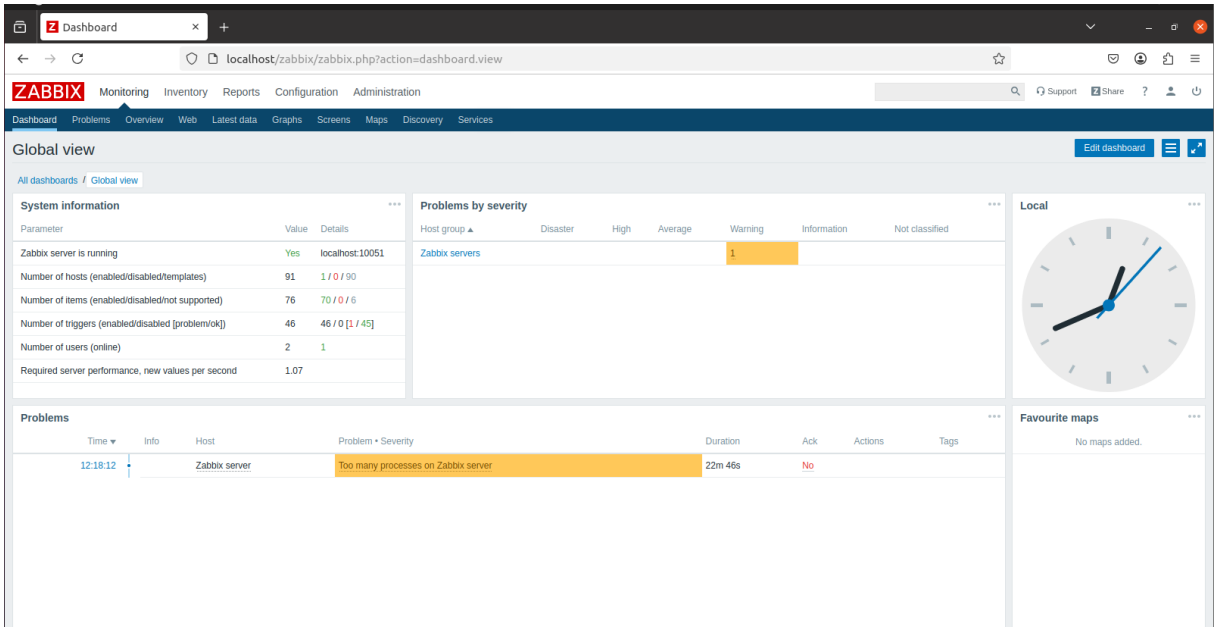


Run: sudo chmod 777 /etc/zabbix



5. Finish the setup wizard and log in using the default credentials:
    - **Username**: Admin (Case-sensitive)
    - **Password**: zabbix





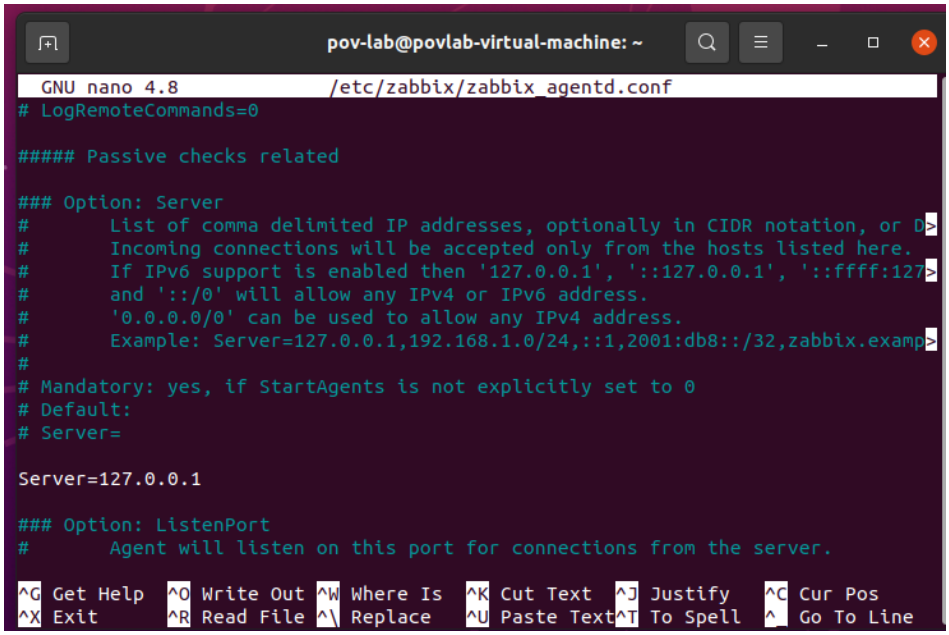
Step 11: Configure Zabbix Agent

Open the agent configuration file:

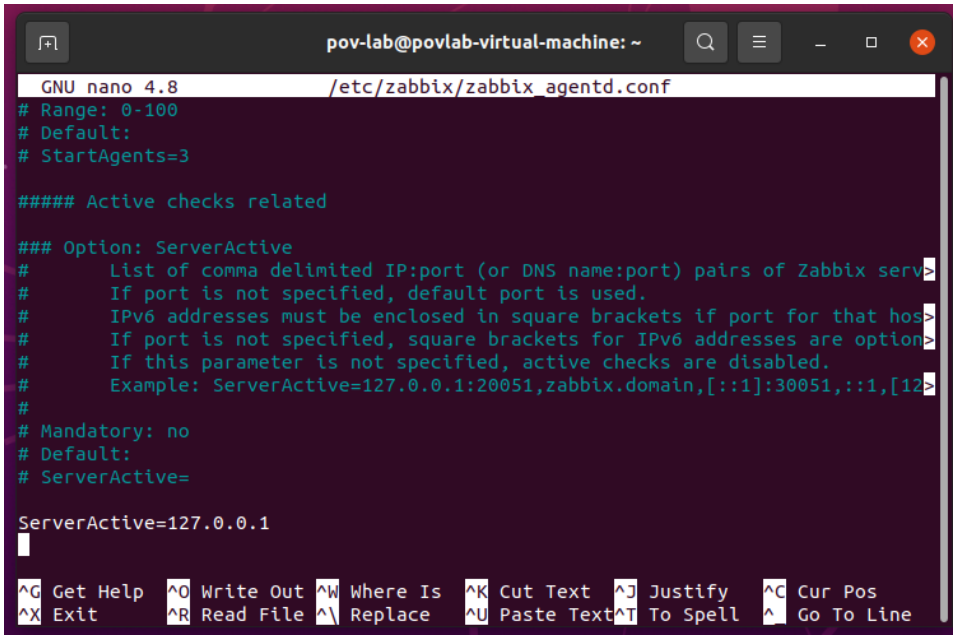
sudo nano /etc/zabbix/zabbix_agentd.conf

Configure the following parameters:

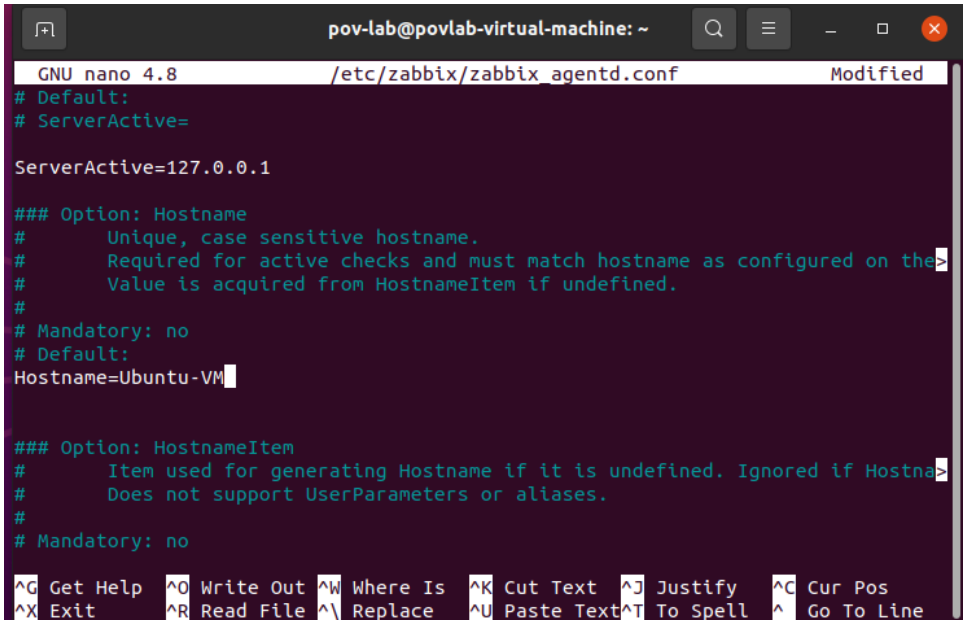
Server=127.0.0.1



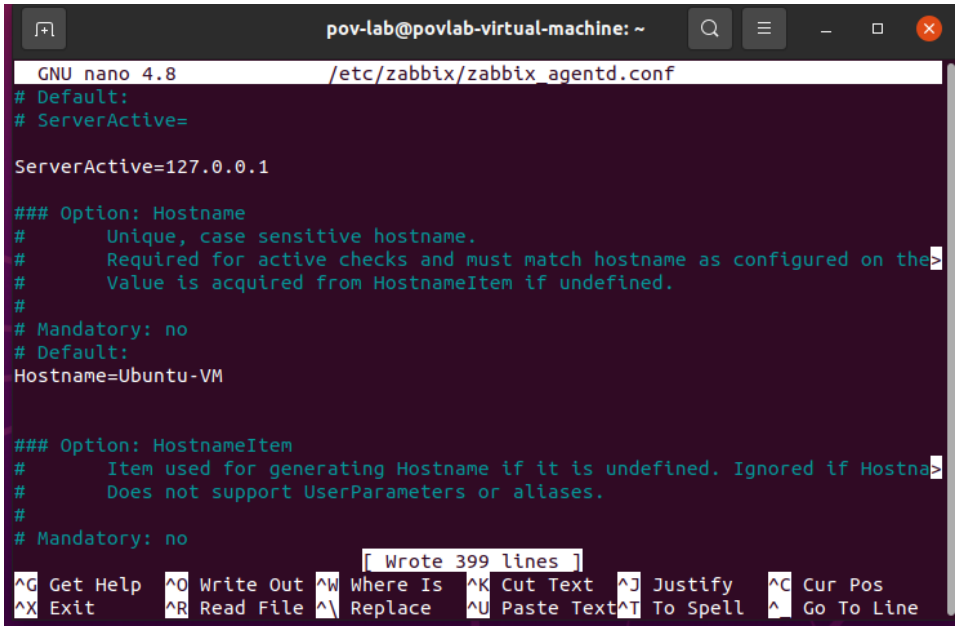
ServerActive=127.0.0.1



Hostname=Ubuntu-VM



Save the file.



Restart the agent:

sudo systemctl restart zabbix-agent

Step 12: Add Ubuntu VM for Monitoring

1. Open the Zabbix web interface.
2. Navigate to Configuration → Hosts.
3. Click Create host.
4. Enter the following details:

Setting	Value
Host name	Ubuntu-VM
Host group	Linux servers
Interface	Agent
IP address	127.0.0.1
Port	10050

5. Select the Template → Template OS Linux
6. Click Add & then click Update.

The Ubuntu virtual machine will now be registered for monitoring.

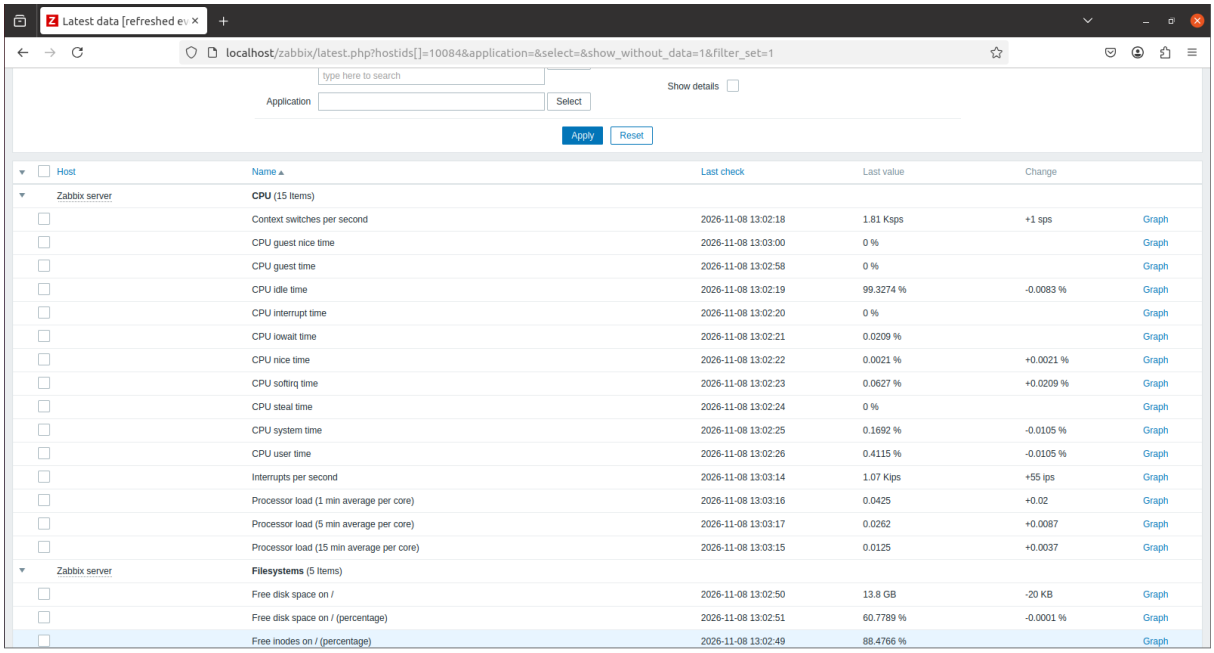
Step 13: Monitor VM Performance

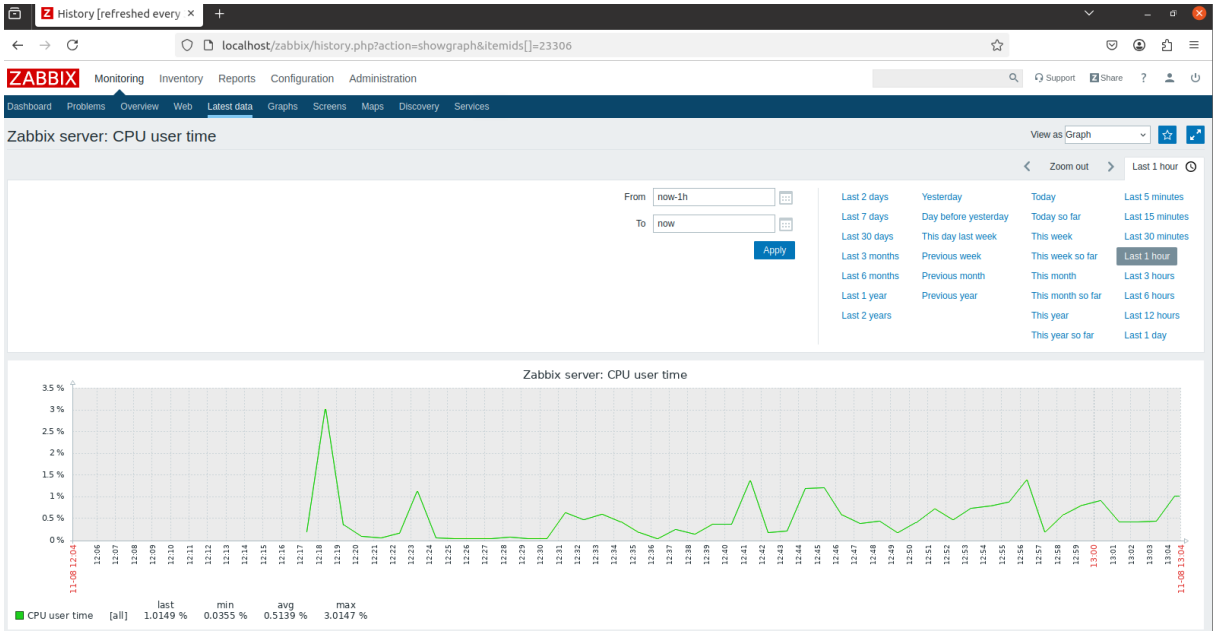
Navigate to the monitoring section of the Zabbix dashboard.

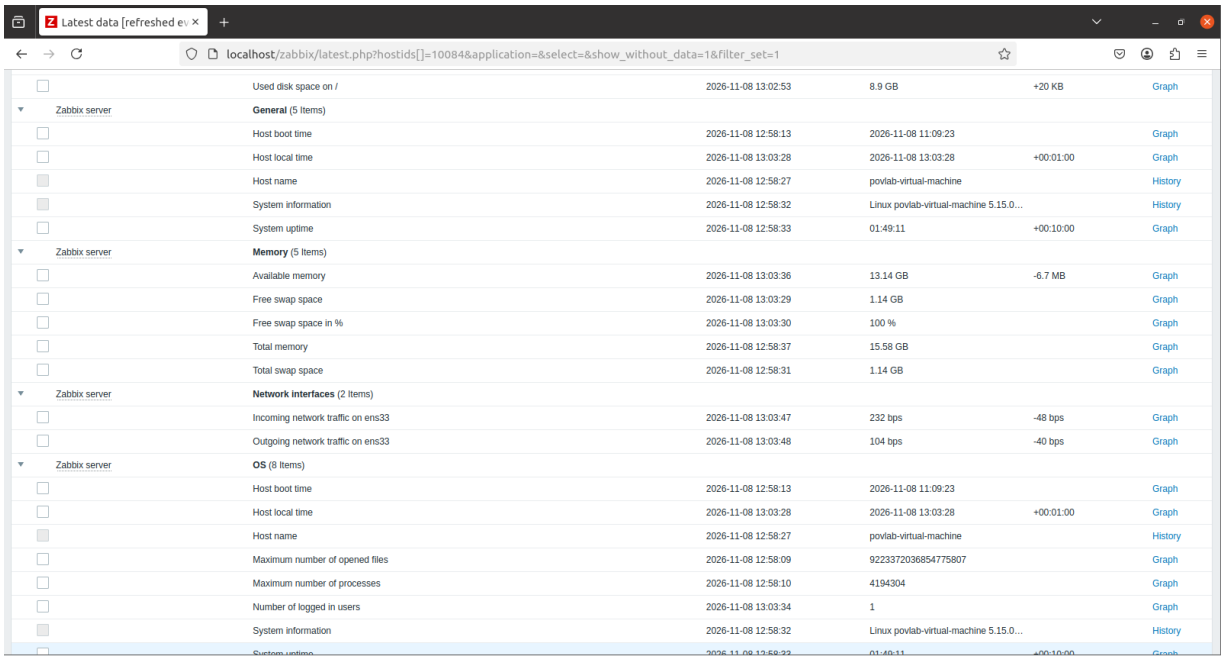
Monitoring → Latest Data → select your host by typing the host name.

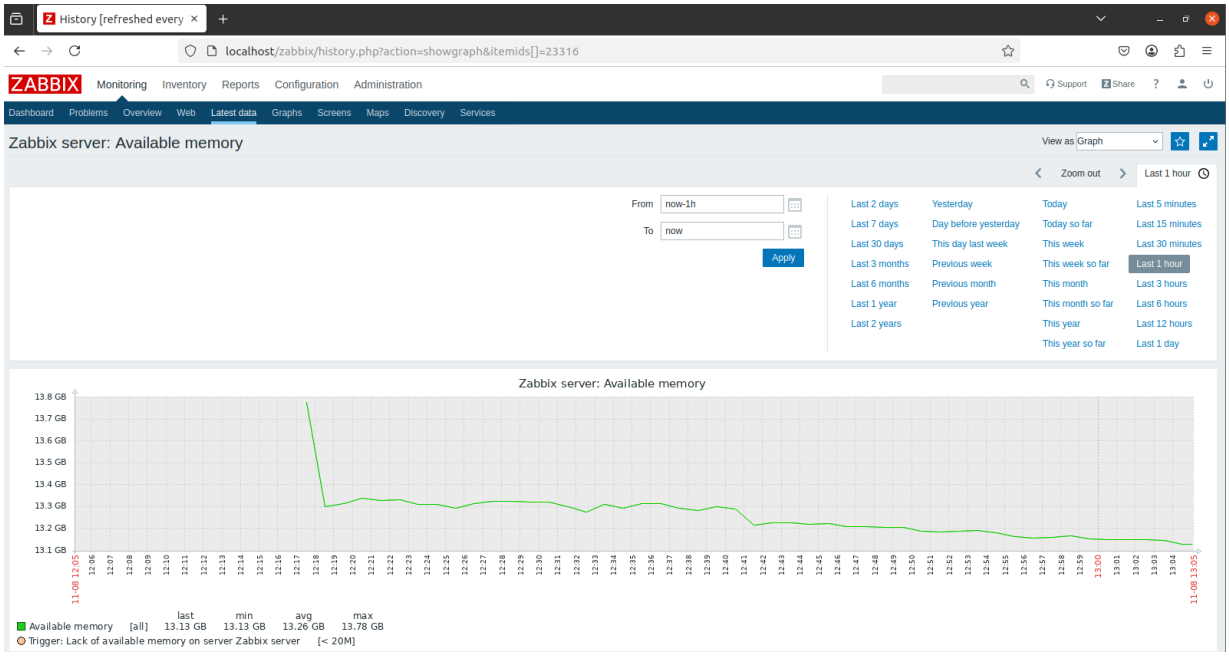
Observe the following parameters:

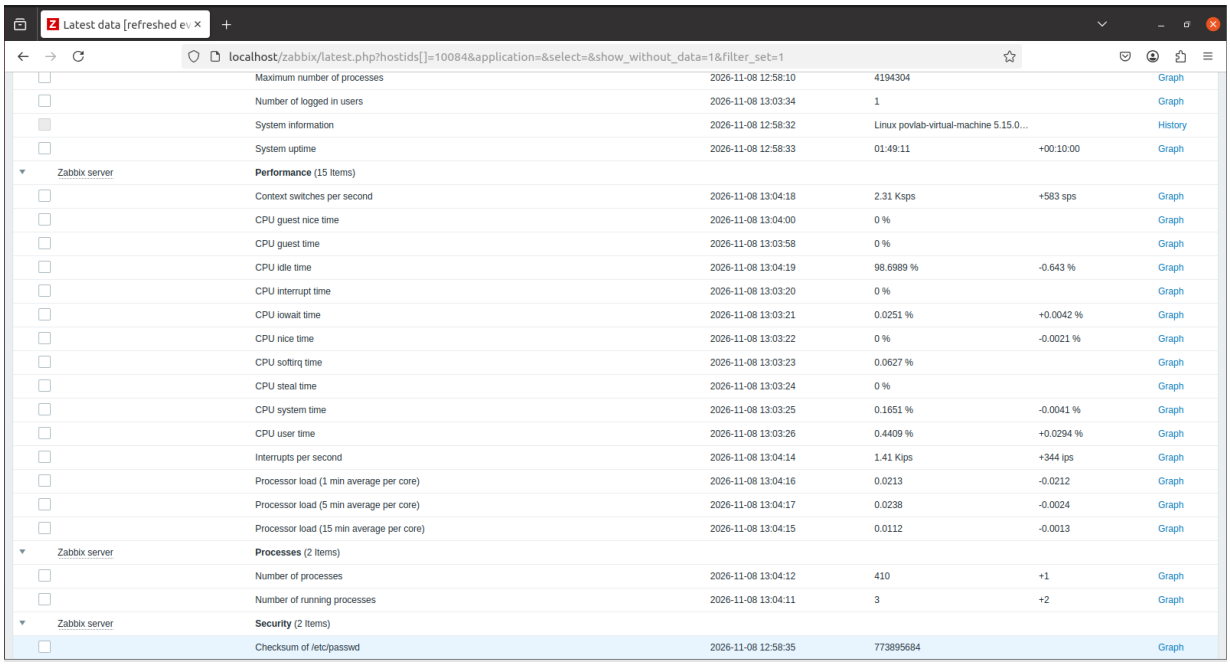
- CPU utilization.
- Memory utilization.
- Disk space usage.
- Network traffic.
- System uptime.
- Agent availability.




History [refreshed every
localhost/zabbix/history.php?action=showgraph&itemids[]=23306
ZABBIX
Monitoring
Inventory
Reports
Configuration
Administration
Support
Share
Dashboard
Problems
Overview
Web
Latest data
Graphs
Screens
Maps
Discovery
Services
Zabbix server: CPU user time
View as
Graph
Zoom out
Last 1 hour
From
now-1h
To
now
Apply
Last 2 days
Yesterday
Today
Last 5 minutes
Last 7 days
Day before yesterday
Today so far
Last 15 minutes
Last 30 days
This day last week
This week
Last 30 minutes
Last 3 months
Previous week
This week so far
Last 1 hour
Last 6 months
Previous month
This month
Last 3 hours
Last 1 year
Previous year
This month so far
Last 6 hours
Last 2 years
This year
Last 12 hours
This year so far
Last 1 day
Zabbix server: CPU user time
3.5 %
3 %
2.5 %
2 %
1.5 %
1 %
0.5 %
0 %
11-08 12:04
11-08 13:04
CPU user time
[all]
last
1.0149 %
min
0.0355 %
avg
0.5139 %
max
3.0147 %


Latest data [refreshed ev
localhost/zabbix/latest.php?hostids[]=10084&application=&select=&show_without_data=1&filter_set=1
Used disk space on / 2026-11-08 13:02:53 8.9 GB +20 KB Graph
Zabbix server General (5 Items)
Host boot time 2026-11-08 12:58:13 2026-11-08 11:09:23 Graph
Host local time 2026-11-08 13:03:28 2026-11-08 13:03:28 +00:01:00 Graph
Host name 2026-11-08 12:58:27 povlab-virtual-machine History
System information 2026-11-08 12:58:32 Linux povlab-virtual-machine 5.15.0… History
System uptime 2026-11-08 12:58:33 01:49:11 +00:10:00 Graph
Zabbix server Memory (5 Items)
Available memory 2026-11-08 13:03:36 13.14 GB -6.7 MB Graph
Free swap space 2026-11-08 13:03:29 1.14 GB Graph
Free swap space in % 2026-11-08 13:03:30 100 % Graph
Total memory 2026-11-08 12:58:37 15.58 GB Graph
Total swap space 2026-11-08 12:58:31 1.14 GB Graph
Zabbix server Network interfaces (2 Items)
Incoming network traffic on ens33 2026-11-08 13:03:47 232 bps -48 bps Graph
Outgoing network traffic on ens33 2026-11-08 13:03:48 104 bps -40 bps Graph
Zabbix server OS (8 Items)
Host boot time 2026-11-08 12:58:13 2026-11-08 11:09:23 Graph
Host local time 2026-11-08 13:03:28 2026-11-08 13:03:28 +00:01:00 Graph
Host name 2026-11-08 12:58:27 povlab-virtual-machine History
Maximum number of opened files 2026-11-08 12:58:09 9223372036854775807 Graph
Maximum number of processes 2026-11-08 12:58:10 4194304 Graph
Number of logged in users 2026-11-08 13:03:34 1 Graph
System information 2026-11-08 12:58:32 Linux povlab-virtual-machine 5.15.0… History





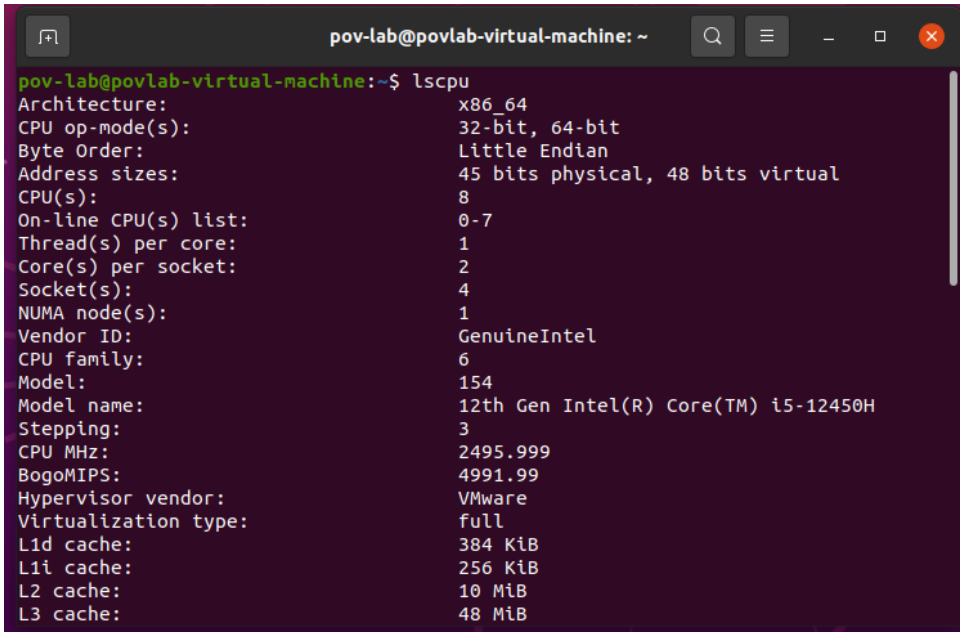
The dashboard displays collected performance data and graphs that help identify resource usage and performance issues.

5. Verification Commands

Execute the following commands in Ubuntu to verify the performance information.

CPU information:

lscpu

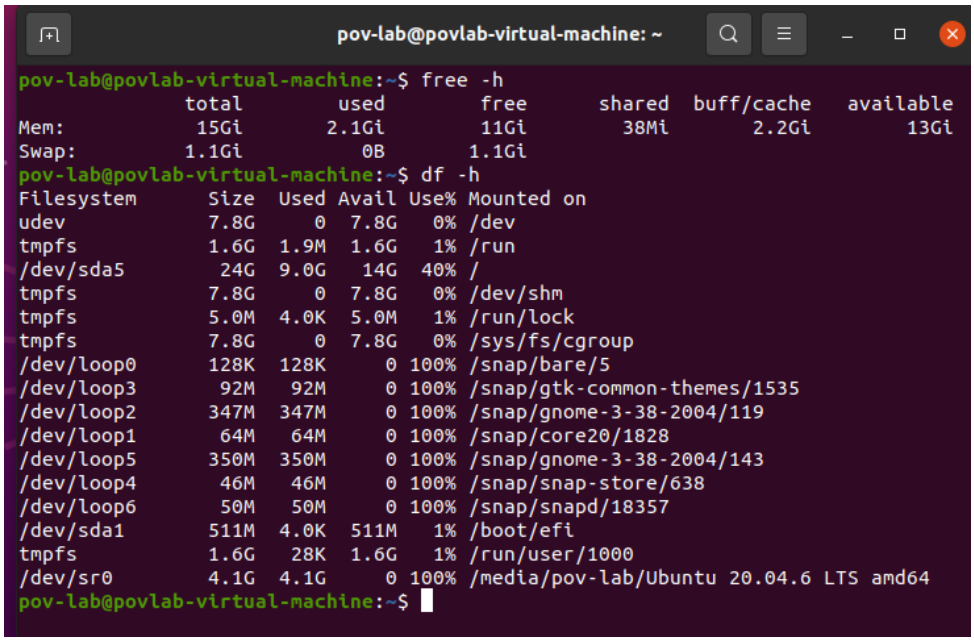


Memory utilization:

free -h

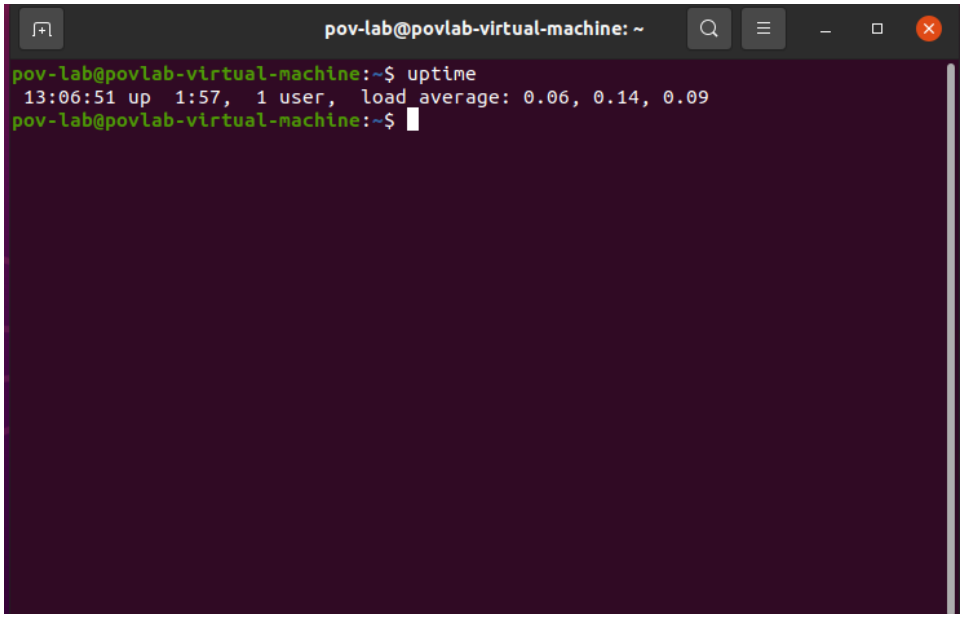
Disk utilization:

df -h



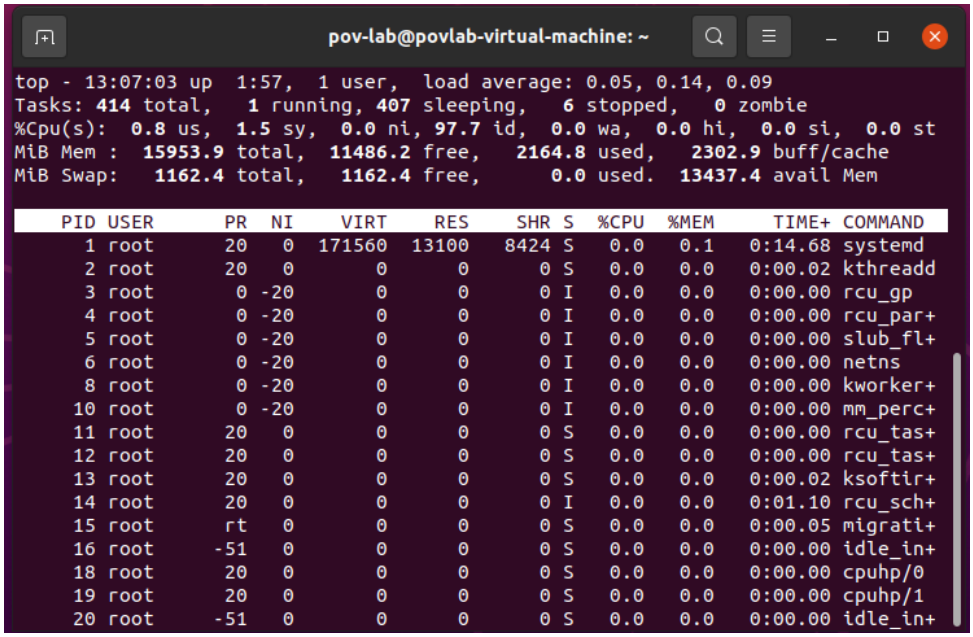
System uptime:

uptime



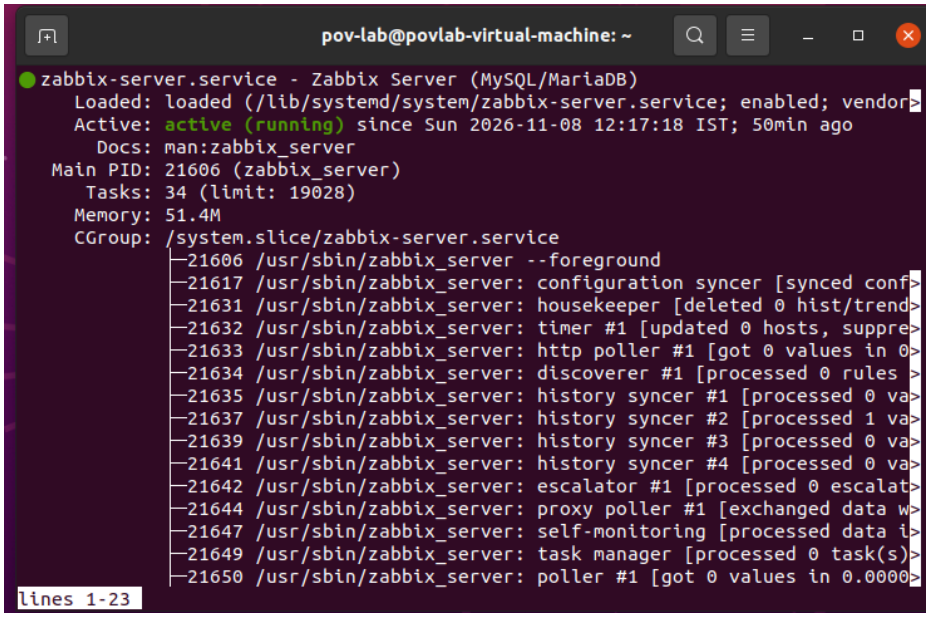
Running processes:

top

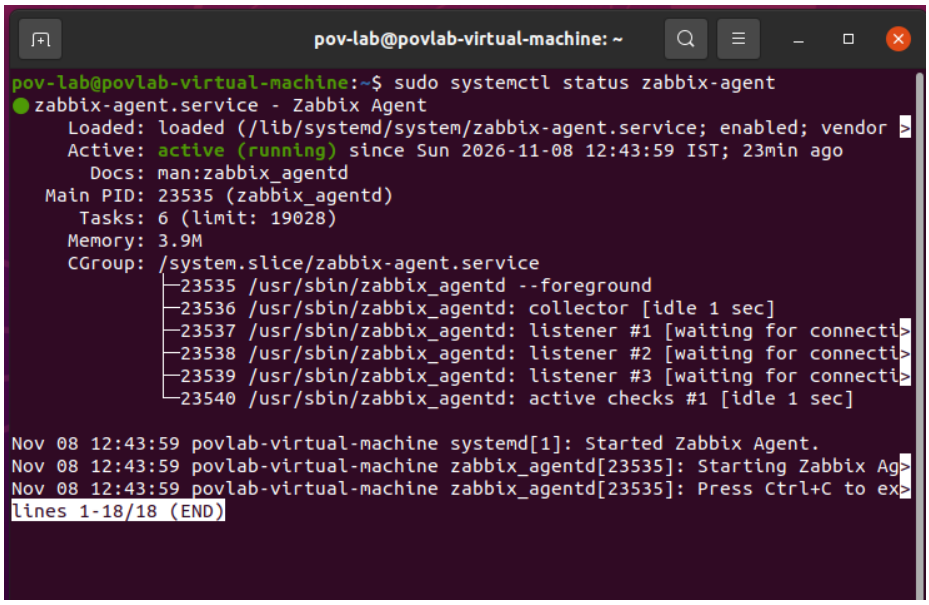


Zabbix service status:

sudo systemctl status zabbix-server



sudo systemctl status zabbix-agent



7. Result

Thus, Zabbix 4.0 LTS was installed and configured successfully on Ubuntu 20.04.6 LTS running in VMware Workstation Pro.

The Zabbix monitoring dashboard was used to monitor important virtual machine performance parameters, including CPU utilization, memory usage, disk space, network traffic and system uptime.

8. Conclusion

Zabbix is an effective open-source VM performance monitoring tool. It provides a centralized dashboard for collecting, analyzing and displaying system performance information.

Through this practical, we learned how to install Zabbix Server and Agent, configure the MySQL database, and monitor the performance of an Ubuntu virtual machine.